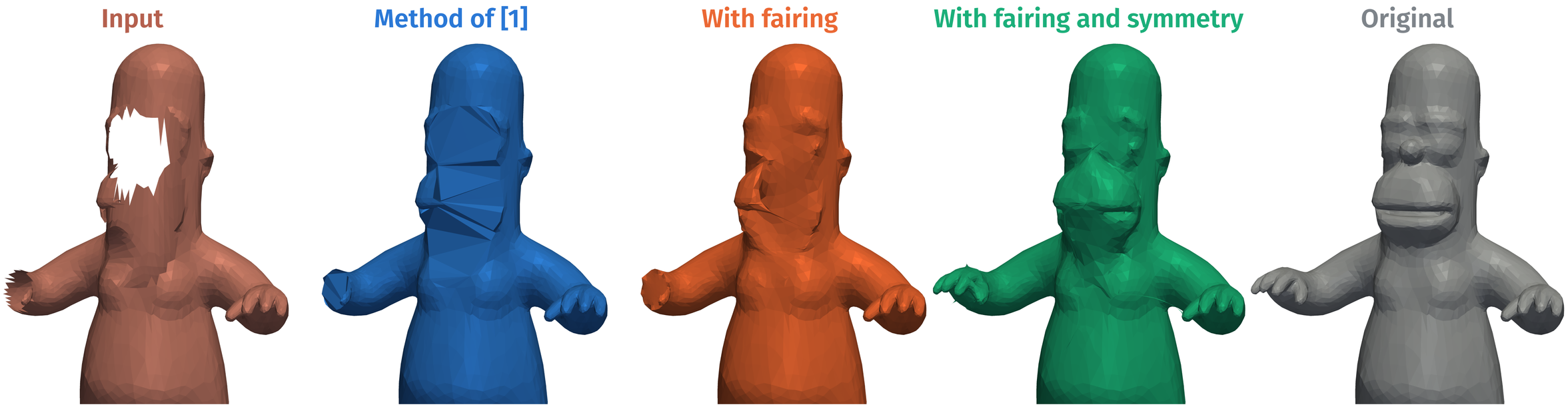
Completing broken meshes

An extension of Jacobson, Kavan and Sorkine-Hornung (SIGGRAPH 2013) with cap fairing and symmetry completion

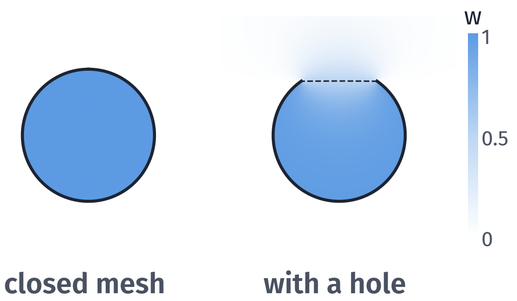
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1 Introduction

Many real meshes have holes, overlapping pieces or loose faces, and simulating or printing a model requires knowing which points are inside. Jacobson et al. [1] use the generalized winding number, which adds up the solid angles that the faces subtend as seen from p.

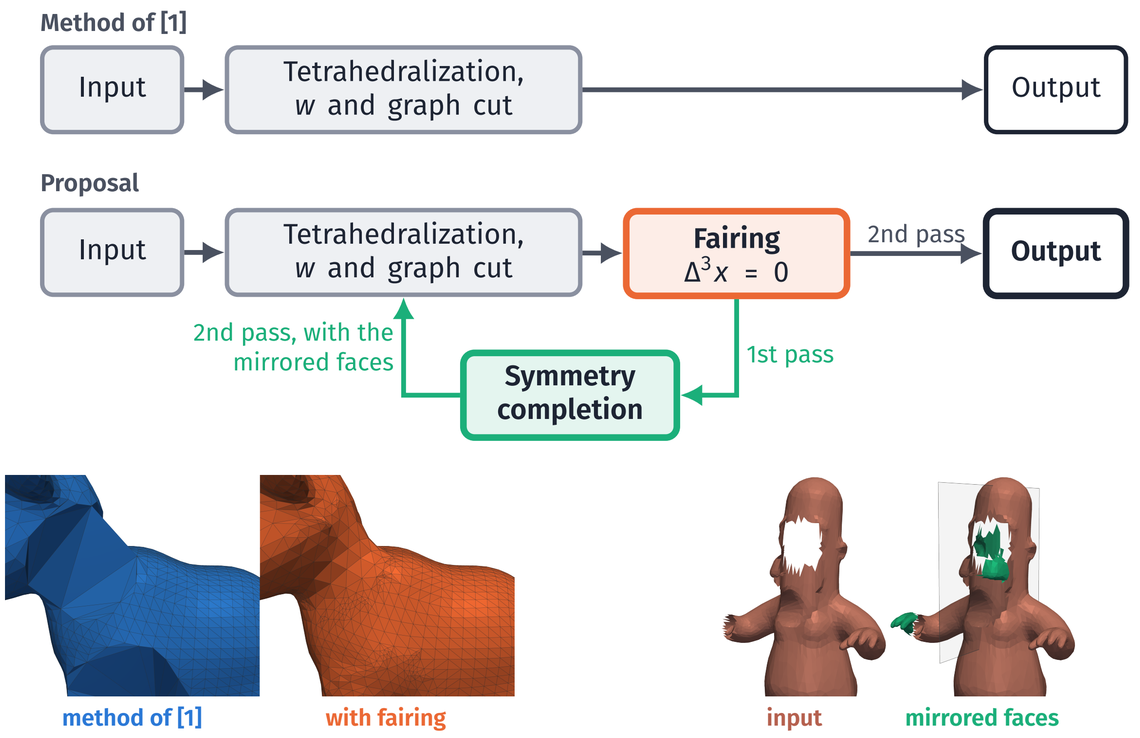
$$w(p) = \frac{1}{4\pi} \sum_f \Omega_f(p)$$



Space is tetrahedralized respecting the input faces [2], w is evaluated in each tetrahedron, and a graph cut chooses which ones lie inside, penalizing the separation of similar neighbors. Two heuristics correct the result so that the output, the boundary of that region, contains the input faces and is a manifold.

2 Proposal

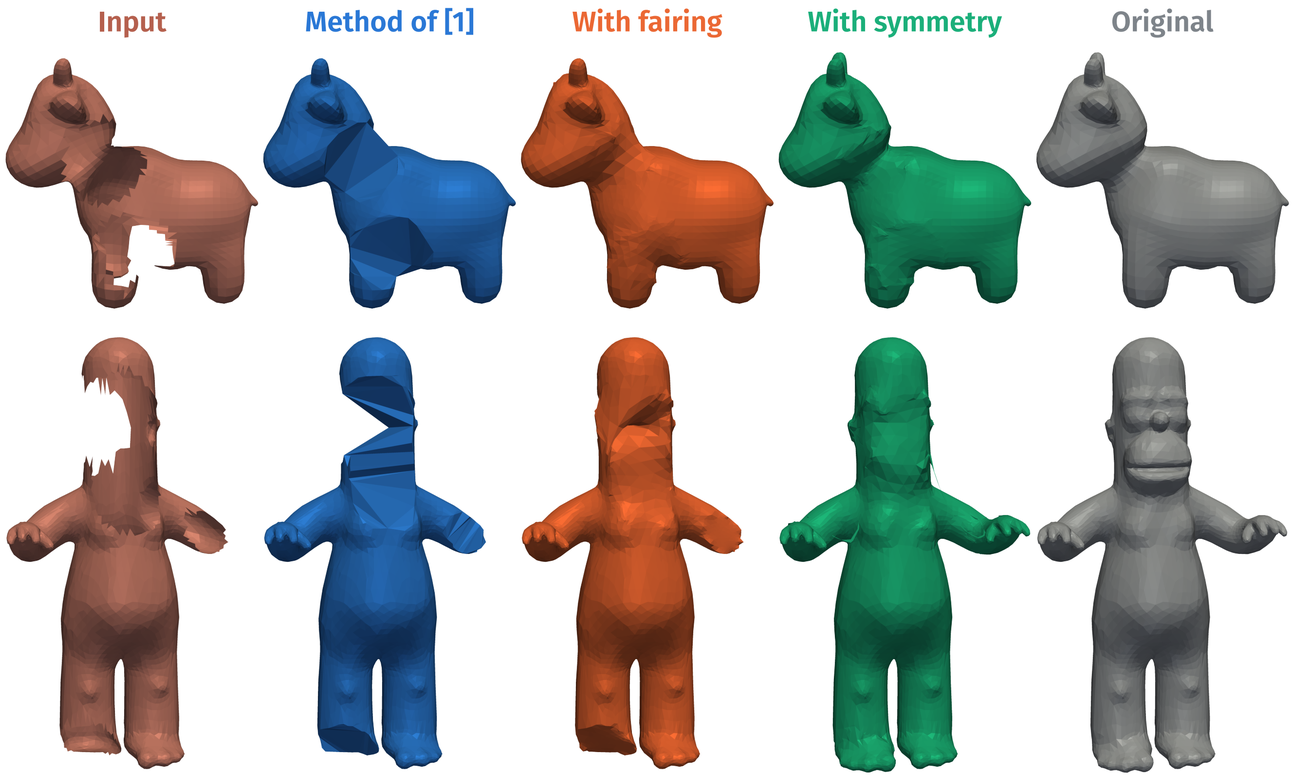
Where surface was missing, the output of [1] is made of faces of the tetrahedralization, which close the hole with flat triangles and no detail. Two stages are added without modifying those of the method, taking ideas from the fairing of TetWild [3] and from symmetry completion [5].



Fairing. The input stays fixed and the cap solves $\Delta^3 x = 0$ [4], which continues it in position, tangent and curvature.

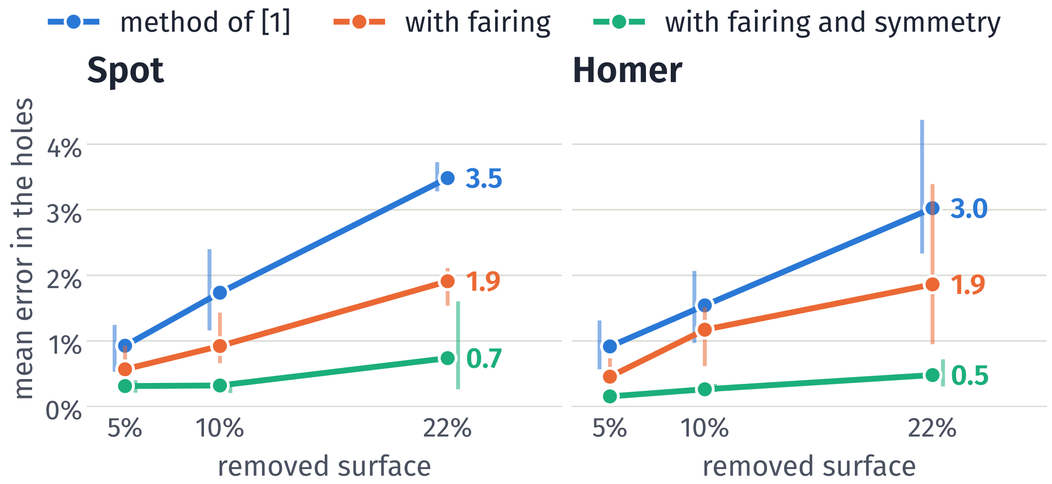
Symmetry. The symmetry plane is estimated, and the mirrored faces that fall in the holes are added.

3 Results



−43%
mean error in the holes with fairing. It improves all 18 cases.

−79%
with fairing and symmetry. It also improves all 18 cases.



4 Limitations

- Symmetry requires a symmetric model whose other side is intact. If surface is missing on both sides of the plane, as with the face of Homer at 22%, there is nothing to copy from.
- Without symmetry, fairing completes the overall shape but not the small details.
- A piece that a hole separates from the rest stays separated, because the cut does not control the topology.

5 Conclusions

- The method of [1] decides correctly what is inside, but it does not give shape to what is missing. A local fairing of the caps, which does not move the input, reduces the error by 43%.
- Symmetry restores details that fairing cannot invent, such as the face and the hand of Homer.
- Future work includes partial symmetries, by parts of the model, and a graph cut that controls the topology.

6 References

[1] A. Jacobson, L. Kavan, O. Sorkine-Hornung. Robust inside-outside segmentation using generalized winding numbers. *ACM TOG* 32(4), 2013.
[2] H. Si. TetGen, a Delaunay-based quality tetrahedral mesh generator. *ACM TOMS* 41(2), 2015.
[3] Y. Hu et al. Tetrahedral meshing in the wild. *ACM TOG* 37(4), 2018.
[4] M. Botsch, L. Kobbelt. An intuitive framework for real-time freeform modeling. *ACM TOG* 23(3), 2004.
[5] X. Li et al. Symmetry and template guided completion of damaged skulls. *Computers & Graphics* 35(4), 2011.

Models: Spot (K. Crane) and Homer (common-3d-test-models). Images and measurements by the author.